Mathematics for Social Sciences

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Out lines

1. **Propositional Logic and Set Theory**
   - Logical connectives
   - Compound (or complex) propositions
   - Tautology and contradiction
2. **Open propositions and quantifiers**
   - Quantifiers
   - Argument and Validity
3. **The concept of a set**

Propositional Logic and Set Theory

Definition

A proposition (or statement) is a sentence which has a truth value (either True or False but not both).

The following sentences are propositions.

(a) Athlet Haile Gebresilasie is an Ethiopian.

(b) 2+3=7

The following sentences are not propositions.

(a) Shut up.

(b) x+3=7

Every proposition has a truth value, namely true (denoted by T) or false (denoted by F). Propositions are denoted by small letters such as p, q, r, and so on.

Logical connectives

(a) Conjunction

When two propositions are joined with the connective "and," the proposition formed is a logical conjunction. "and" is denoted by "$\wedge$". So, the logical conjunction of two propositions, p and q, is written as:
$p \wedge q$ read as " p and q," or " p conjunction q ". p and q are called the components of the conjunction. $p \wedge q$ is true if and only if p is true and q is true.

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

Example

Consider the following propositions:

p: 3 is an odd number. (True)

q: 27 is a prime number. (False)

r: Addis Ababa is the capital city of Ethiopia. (True)

$p \wedge q$: 3 is an odd number and 27 is a prime number. (False)

$p \wedge r$: 3 is an odd number and Addis Ababa is the capital city of Ethiopia. (True)

Disjunction

(b) disjunction

When two propositions are joined with the connective "or," the proposition formed is a logical conjunction. "and" is denoted by "$\vee$". So, the logical disnjunction of two propositions, p and q, is written as:
$p \vee q$ read as " p or q," or " p disjunction q ". p and q are called the components of the conjunction. $p \vee q$ is false if and only if p is false and q is false.

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

Example

Consider the following propositions:

p: 3 is an odd number. (True)

q: 27 is a prime number. (False)

r: Addis Ababa is the capital city of Ethiopia. (True)

$p \vee q$: 3 is an odd number or 27 is a prime number. (True)

$p \vee r$: 3 is an odd number or Addis Ababa is the capital city of Ethiopia. (True)

Implication

(c) Implication

When two propositions are joined with the connective "implies," the proposition formed is a logical implication. "implies" is denoted by " $\Longrightarrow$ ". So, the logical implication of two propositions, p and q, is written as:
$p \Longrightarrow q$ read as " p implies q," $p \Longrightarrow q$ is false if and only if p is true and q is false.

p	q	$p \Longrightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

Example

Consider the following propositions:
p: 3 is an odd number. (True)
q: 27 is a prime number. (False)
r: Addis Ababa is the capital city of Ethiopia. (True)
$p \implies q$: 3 is an odd number implies 27 is a prime number. (False)
$p \implies r$: 3 is an odd number implies Addis Ababa is the capital city of Ethiopia. (True)

$p \implies q$ can also be expressed as follows.

- If p, then q.
- q if p.
- p only if q.
- p is sufficient for q.
- q is necessary for p

Bi-implication

(d) Bi-implication

When two propositions are joined with the connective "bi-implies," the proposition formed is a logical bi-implication. "Bi-implication" is denoted by "$\Leftrightarrow$". So, the logical bi-implication of two propositions, p and q, is written as:
$p \Leftrightarrow q$ read as " p bi-implies q," $p \Leftrightarrow$ is false if and only if p and q have different truth values.

p	q	$p \Leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

Example

Consider the following propositions:
p: 3 is an odd number. (True)
q: 27 is a prime number. (False)
r: Addis Ababa is the capital city of Ethiopia. (True)
$p \Leftrightarrow q$: 3 is an odd number bi-implies 27 is a prime number. (False)
$p \Leftrightarrow r$: 3 is an odd number bi-implies Addis Ababa is the capital city of Ethiopia. (True)

$p \Leftrightarrow q$ can also be expressed as follows.

- p if and only if q.
- p is sufficient and necessary for q.
- q is sufficient and necessary for p.
- p implies q and q implies p
- p is equivalent to q.

Negation

Given any proposition p, we can form the proposition $\neg p$ called the negation of p. The truth value of $\neg p$ is F if p is T and T if p is F.

Example

Let p: Addis Ababa is the capital city of Ethiopia. (True)
$\neg p$: Addis Ababa is not the capital city of Ethiopia. (False)

Compound (or complex) propositions

Definition

The proposition formed by joining two or more proposition by connective(s) is called a compound statement.

Example

$p \wedge (q \implies r)$, $p \wedge (\neg q \vee r)$

The possible truth values of a proposition are often listed in a table, called a truth table. If the compound proposition contains two component propositions, then there are four possible combinations of truth values for the components. If the compound proposition contains three component propositions, then there are eight possible combinations of truth values for the components. In general, if the compound proposition contains n component propositions, then there are 2^n possible combinations of truth values for the components.

Definition

Two compound propositions P and Q are said to be equivalent if they have the same truth value for all possible combinations of truth values for the component propositions occurring in both P and Q. In this case we write $P \equiv Q$.

Example

Show that $P : p \implies q$ and $Q : \neg q \implies \neg p$ are equivalent.

Given the conditional $p \implies q$.

- $q \implies p$ is the converse of $p \implies q$.
- $\neg p \implies \neg q$ is the inverse of $p \implies q$,
- $\neg q \implies \neg p$ is the contapositive of $p \implies q$.

Example

If Kidist lives in Addis Ababa, then she lives in Ethiopia.
Converse: If Kidist lives in Ethiopia, then she lives in Addis Ababa.
Contrapositive: If Kidist does not live in Ethiopia, then she does not live in Addis Ababa.
Inverse: If Kidist does not live in Addis Ababa, then she does not live in Ethiopia.

Tautology and contradiction

Definition

A compound proposition is a tautology if it is always true regardless of the truth values of its component propositions. If, on the other hand, a compound proposition is always false regardless of its component propositions, we say that such a proposition is a contradiction.

A proposition that is neither a tautology nor a contradiction is called a contingency.

Example

$p \vee \neg p$ is tautology, $p \wedge \neg p$ is contradiction, $p \implies q$ is contingency

Logical equivalence, which satisfy various laws or identities are presented below.

1. Idempotent Laws:
   (a) $p \equiv p \wedge p$,
   (b) $p \equiv p \vee p$
2. Commutative Laws:
   (a) $p \wedge q \equiv q \wedge p$,
   (b) $p \vee q \equiv q \vee p$
3. Associative Laws;
   (a) $p \wedge (q \wedge r) \equiv (p \wedge q) \wedge r$,
   (b) $p \vee (q \vee r) \equiv (p \vee q) \vee r$
4. Distributive Laws:
   (a) $p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$,
   (b) $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$

5. De Morgan's Laws:
   (a) $\neg(p \vee q) \equiv \neg p \wedge \neg q$,
   (b) $\neg(p \wedge q) \equiv \neg p \vee \neg q$
6. Law of Contrapositive:
   $p \implies q \equiv \neg q \implies \neg p$.
7. Complement Law:
   $\neg(\neg p) \equiv p$.

Open propositions and quantifiers

Definition

An open statement (also called a predicate) is a sentence that contains one or more variables and whose truth value depends on the values assigned for the variables.

We represent an open statement by a capital letter followed by the variable(s) in parenthesis, example $P(x)$, $Q(x, y)$ etc.

Example

Here are some open propositions:
x is the day before Sunday.
x is a city in Africa.
x is greater than y

The collection of all allowable values for the variable in an open sentence is called the **universal set** (**the universe** of discourse) and denoted by **U**

Definition

Two open propositions $P(x)$ and $Q(x)$ are said to be equivalent if and only if $P(a) \equiv Q(a)$ for all individual a in the specified (universal) set U..

Example

Let $P(x) : x^2 - 1 = 0$
$Q(x) : |x| \geq 1$. Show that $P(x)$ and $Q(x)$ are equivalent in $U = \{-1, -\frac{1}{2}, 0, , 1\}$.

Quantifiers

1. Universal quantifier:
   - The phrases "for every x", "for each x" and "for all x" are universal quantifiers.
   - denoted by $\forall x$,
   - If $P(x)$ is an open proposition with universe U, then $\forall x(P(x)$, is a quantified proposition and is read as "for every $x \in U$ with property $P(x)$.
2. Existential quantifier:
   - The phrases "for some x", "for at least one x" and "there exists an x" are existential quantifiers.
   - denoted by $\exists x$,
   - If $P(x)$ is an open proposition with universe U, then $\exists x(P(x)$, is a quantified proposition and is read as "for there exists" $x \in U$ with property $P(x)$.

To show that $\forall x(P(x))$ is F, it is sufficient to find at least one $a \in U$ such that $P(a)$ is False.
To show that $\exists x(P(x))$ is T, it is sufficient to find at least one $a \in U$ such that $P(a)$ is True.

Example

Write the following statements using quantifiers, and determine the truth value.

(a) For each real number $x > 0$ such that $x^2 + 1 > 0$

(b) There is a real number x such that $x^2 + 5x + 6 = 0$

(c) he square of any real number is nonnegative.

Negation of quantified proposition

- $\neg(\forall x)P(x) \equiv (\exists x)\neg P(x)$
- $\neg(\exists x)P(x) \equiv (\forall x)\neg P(x)$

Example

Let $U = \mathbb{R}$.
$\neg(\exists x)(x < x^2) \equiv (\forall x)(x \geq x^2)$

Example

1. Symbolize (translate) the following statements involving quantifiers.
   - **(a)** All rationals are reals.
   - **(b)** No rationals are reals.
   - **(c)** Some rationals are reals.
   - **(d)** Some rationals are not reals.
2. Let $U =$The set of integers, $P(x)$: x is a prime number, $Q(x)$: x is an even number, $R(x)$: x is an odd number. Determine the truth value of the following propositions.
   - **(a)** $\exists x(P(x) \implies Q(x))$
   - **(b)** $\forall x(P(x) \implies Q(x))$
   - **(c)** $\forall x(R(x) \wedge Q(x))$

Quantifiers Occurring in Combinations

(i) $(\forall x)(\forall y)P(x, y)$ means "for all x and for all y property $P(x, y)$ holds.

(ii) $(\exists x)(\exists y)P(x, y)$ means "there exists an x and there exists a y for which $P(x, y)$ holds.

(iii) $(\forall x)(\exists y)P(x, y)$ means "for every x there exists a y such that $P(x, y)$ holds.

(iv) $(\exists x)(\forall y)P(x, y)$ means "there exists an x which stands to every y with relation $P(x, y)$.

Example

Determine the truth value of the following.
Let $U =$ The set of integers.

(i) $(\forall x)(\forall y)(x + y = 7)$

(ii) $(\exists x)(\exists y)(x + y = 7)$

(iii) $(\forall x)(\exists y)(x + y = 7)$

(iv) $(\exists x)(\forall y)(x + y = 7)$

Argument and Validity

Definition

An argument (logical deduction) is an assertion that a given set of statements $p_1, p_2, \cdots, p_n$, called hypotheses or premises, yield another statement q, called the conclusion.

Such a logical deduction is denoted by: $p_1, p_2, \cdots, p_n \vdash q$ or

$$\begin{array}{c} p_1 \\ p_2 \\ . \\ . \\ . \\ p_n \\ \hline q \end{array}$$

Example

If Addis Ababa is in Ethiopia, then it is in Africa. Addis Ababa is not in Africa. Therefore, Addis Ababa is not in Ethiopia. Write the argument form.

Definition

An argument form $p_1, p_2, \cdots, p_n \vdash q$ is said to be **valid** if q is true whenever all the premises $p_1, p_2, \cdots, p_n$ are true; otherwise it is **invalid**.

Example

Investigate the validity of the following argument:

(a) $p \implies q, \neg q \vdash \neg p$

(b) $p \implies q, \neg q \implies r \vdash p$

Rules of inferences

1. Modes Ponens:

$$\begin{array}{l} p \\ p \implies q \\ \hline q \end{array}$$

2. Modes Tollens:

$$\begin{array}{l} \neg q \\ p \implies q \\ \hline \neg p \end{array}$$

3. Principle of Syllogism

$$\begin{array}{l} p \implies q \\ q \implies r \\ \hline p \implies r \end{array}$$

4. Principle of Adjunction:

(a) $$\frac{\begin{array}{l} p \\ q \end{array}}{p \wedge q}$$

(b) $$\frac{p}{p \vee q}$$

5. Principle of Detachment:

$$\frac{p \wedge q}{p, q}$$

6. Modes Tollendo Ponens:

$$\frac{\begin{array}{l} \neg p \\ p \vee q \end{array}}{q}$$

7. Modes Ponendo Tollens:

$$\frac{\begin{array}{l} \neg(p \wedge q) \\ p \end{array}}{\neg q}$$

8. Constructive Dilemma:

$$\frac{\begin{array}{l}(p \implies q) \wedge (r \implies s)\\ p \vee r\end{array}}{q \vee s}$$

9. Principle of Equivalence:

$$\frac{\begin{array}{l}p \Leftrightarrow q\\ p\end{array}}{q}$$

10. Principle of Conditionalization:

$$\frac{p}{q \implies p}$$

The concept of a set

- The term set refers to a well-defined collection of objects that share a certain property or certain properties.
- we use capital letters to denote the names of sets and lowercase letters for elements of a set.
- If A is a set, then the objects of the collection are called the elements or members of the set A. If x is an element of the set A, then we write as $x \in A$. If x is not an element of the set A, then we write as $x \notin A$.

Descriptions of set

Sets are described or characterized by one of the following four different ways.

1. Verbal Method: using an ordinary English statement.

Example

The set of vowels in English alphabet.
The set of all countries in Africa.

2. Roster/Complete Listing Method: listing all elements of a set.

Example

$A = \{a, e, i, o, u\}$

3. Partial Listing Method:when the number of elements of a set may be too large to list them all, we list out few elements followed (or preceded) by three dotes

Example

$\mathbb{N} = \{1, 2, 3, \cdots\}$

4. Set-builder Method: using an open proposition.

The general form is $A = \{x : P(x)\}$ or $A = \{x \mid P(x)\}$

Example

$A = \{x : \text{x is a vowel in English alphabet}\}$
$B = \{n : \text{n is aneven integer}\}$

Definition

The set which has no element is called the empty (or null) set and is denoted by $\emptyset$ or $\{\}$.

Example

$A = \{x \in \mathbb{R} : x^2 + 1 = 0\}$ is an empty set.

Definition

A set is finite if it has limited number of elements and it is called infinite if it has unlimited number of elements.

Relationships between two sets

Definition

A set B is said to be a subset of set A (or is contained in A), denoted by $B \subseteq A$, if every elements of B is is an element of A. That is,

$$(\forall x)(x \in B \implies x \in A)$$

For any set A, $\emptyset \subseteq A$ and $A \subseteq A$.

Example

Let $A = \{1, 2, 3, 4, 5\}$, $B = \{2, 4\}$, $D = \{x \in \mathbb{N} \mid 1 \leq x \leq 5\}$. Thus, $B \subseteq A$, $A \subseteq D$.

Definition

Sets A and B are said to be equal if they contain exactly the same elements. That is,

$$(\forall x)(x \in B \Leftrightarrow x \in A)$$

Example

The sets $\{1, 2, 3\}$, $\{2, 1, 3\}$ and $\{3, 2, 1\}$ are equal.
$A = \{1, 2, 3, 4, 5\}$ and $D = \{x \in \mathbb{N} \mid 1 \leq x \leq 5\}$ are equal.

Definition

Set B is said to be a proper subset of set A if every element of B is also an element of A, but B has at least one element that is not in A. In this case, we write $B \subset A$. That is,

$$B \subset A \Leftrightarrow (\forall x)(x \in B \implies x \in A) \wedge (A \neq B)$$

Definition

Let A be a set. The power set of A, $P(A)$ is the set whose elements are all A. That is

$$P(A) = \{B \mid B \subseteq A\}$$

Example

Find the power set of

(a) $A = \{\emptyset, 0\}$.

(b) $B = \{a, b, c\}$

If a set A is finite with elements n, then the number of subsets of A is 2^n.

Set Operations and Venn diagrams

Definition

The union of two sets A and B, denoted by $A \cup B$,is the set of all elements that are either in A or in B (or in both sets). That is,

$$A \cup B = \{x \mid (x \in A) \vee (x \in B)\}$$

Definition

The intersection of two sets A and B, denoted by $A \cap B$,is the set of all elements that are in A and B . That is,

$$A \cap B = \{x \mid (x \in A) \wedge (x \in B)\}$$

Definition

The difference of two sets A and B, denoted by $A - B$,is the set of all elements in A but not in B . That is,

$$A - B = \{x \mid (x \in A) \wedge (x \notin B)\}$$

Definition

The difference of two sets A and B, denoted by $A - B$, is the set of all elements in A but not in B. This set is also called the relative complement of B with respect to A. Mathematically,

$$A - B = \{x \mid (x \in A) \wedge (x \notin B)\}$$

$A - B$ also denoted by $A \backslash B$.

Definition

: Let A be a subset of a universal set U. The absolute complement (or simply complement) of A, denoted by A' (or A^c or $\bar{A}$), is defined to be the set of all elements of U that are not in A. That is

$$A' = \{x : x \in U \wedge x \notin A\}$$

Definition

The symmetric difference of two sets A and B, denoted by $A \triangle B$ is the set

$$A \triangle B = (A - B) \cup (B - A)$$

Example

Theorem

For any two sets A and B, the following holds.

(a) $(A')' = A$

(b) $A' = U - A$

(c) $A - B = A \cap B'$

(d) $(A \cup B)' = A' \cap B'$, $(A \cap B)' = A' \cup B'$ *(DeMorgan's law)*

(e) $A \subseteq B \implies B' \subseteq A'$

Theorem

For any three sets A, B and C , the following holds.

(a) $A \cup B = B \cup A$ *($\cup$ is commutative)*

(b) $A \cap B = B \cap A$ *($\cap$ is commutative)*

(c) $(A \cup B) \cup C = A \cup (B \cup C)$ *($\cup$ is associative)*

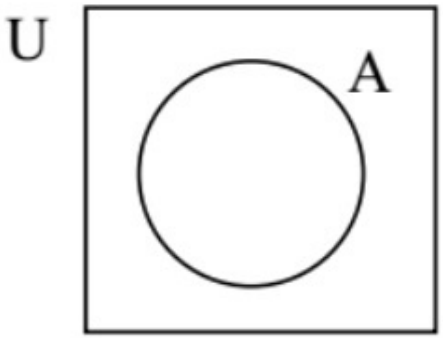
(d) $(A \cap B) \cap C = A \cap (B \cap C)$ *($\cap$ is associative)*

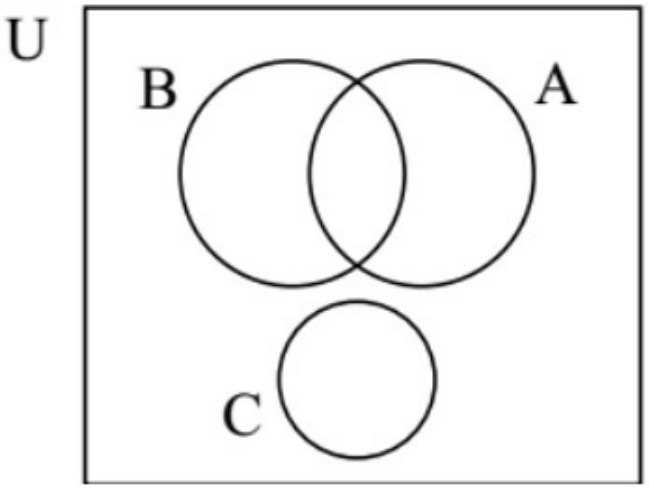
(e) $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$ *($\cup$ is distributive over $\cap$)*

(f) $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$ *($\cap$ is distributive over $\cup$)*

Venn diagrams

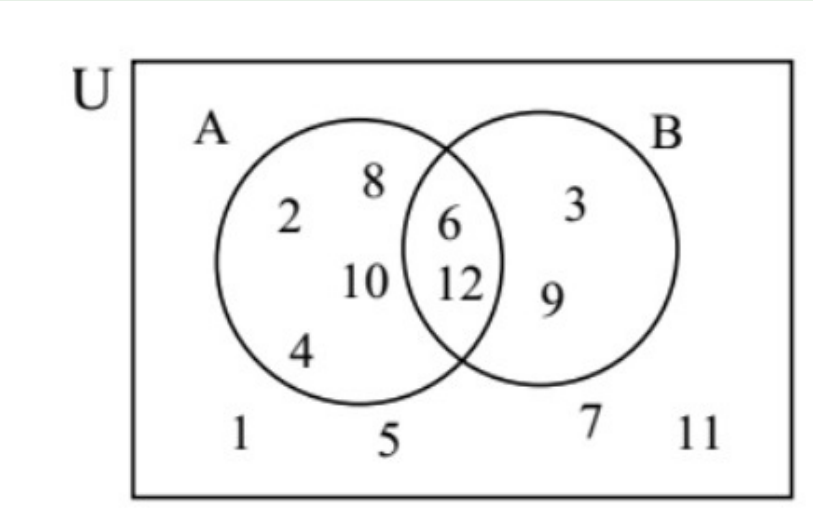
A Venn diagram is a schematic or pictorial representative of the sets involved in the discussion.
Usually sets are represented as interlocking circles, each of which is enclosed in a rectangle, which represents the universal set U.





Example

Consider the venn diagram below, and find $A \cap B$, $A \backslash B$, $A \triangle B$.



Mathematics for Social Sciences

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Out lines

1. **Function**
   - The real number system and arithmetic operations
   - Equations and Inequalities: Linear and Quadratic
   - Review of relations and functions
   - Real Valued functions and their properties
     - Polynomial function
     - Rational Functions and their Graphs
     - Exponential function
     - Logarithm function

The real number system

Definition

The set of natural numbers is given by $\mathbb{N} = \{1, 2, 3, \cdots\}$.
The set of integers is given by
$\mathbb{Z} = \{\cdots, -3, -2, -1, 0, 1, 2, 3, \cdots\}$.
The set of rational numbers is given by
$\mathbb{Q} = \left\{\frac{a}{b} : a \in \mathbb{Z} \wedge 0 \neq b \in \mathbb{Z}\right\}$.

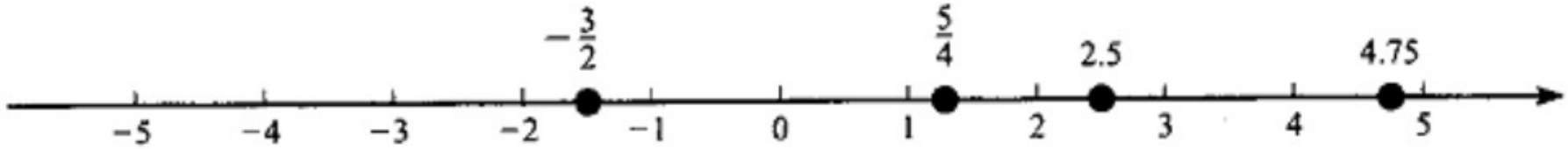
A number which can not be written as the quotient of two integers is an irrational number.

Definition

The set of real numbers denoted by $\mathbb{R}$ can be described as the union of the set of rational and irrational numbers. That is,

$$\mathbb{R} = \{x : x \text{ is a rational number or an irrational number}\}$$

Each point on the number line corresponds a unique real number and vice-versa.



The four arithmetic operations

Addition, subtraction, multiplication and Division.

Properties

- The commutative Properties:
  1. For addition: $a + b = b + a$
  2. For multiplication: $ab = ba$

- The associative properties:

  3. For addition: $a + (b + c) = (a + b) + c$
  4. For multiplication: $a(bc) = (ab)c$

- The distributive properties:

  5. $a(b + c) = ab + ac$ or $(b + c)a = ba + ca$

- Identities:

  6. For addition: There is a unique number called the additive identity, represented by 0, which has the property that $a + 0 = 0 + a$ for all real numbers a.
  7. For Multiplication: There is a unique number called the multiplicative identity, represented by 1, which has the property that $a.1 = 1.a$ for all real numbers a.

- Inverses:
  8. For addition: Each real number a has a unique additive inverse, represented by $-a$ which has the property that $a + -a = -a + a = 0$
  9. For Multiplication: Each real number a, except 0 has a unique additive inverse, represented by $\frac{1}{a}$ which has the property that $a.\left(\frac{1}{a}\right) = \left(\frac{1}{a}\right).a = 1$
- Closure properties:
  10. For addition: The sum of two real numbers is a real number.
  11. For Multiplication: The product of two real numbers is a real number.

Subtraction and division are defined by:

$x - y = x + (-y)$ and $x \div y = x.\left(\frac{1}{y}\right)$, where $y \neq 0$.

The order relations

The order relation on the set of real numbers: We compare two real numbers using $<$ or $>$ or $\leq$ or $\geq$.

The order property

1. Trichotomy: If x and y are numbers, exactly one of the following holds: $x < y$ or $y < x$ or $x = y$
2. Transitivity: $x < y, y < z \implies x < z$
3. Addition:$x < y \Leftrightarrow x + z < y + z$
4. Multiplication:
   When z is positive, $x < y \Leftrightarrow x.z < y.z$,
   When z is negative, $x < y \Leftrightarrow x.z > y.z$.

Let a and b be two real numbers such that $a < b$, then the intervals which are subsets of $\mathbb{R}$ with end points a and b are denoted and defined as below:

(i) $(a, b) = \{x : a < x < b\}$ open interval from a to b,

(ii) $[a, b] = \{x : a \leq x \leq b\}$ closed interval from a to b,

(iii) $(a, b] = \{x : a < x \leq b\}$ open-closed interval from a to b,

(iv) $[a, b) = \{x : a \leq x < b\}$ closed-open from interval from a to b,

Linear Equations and Inequalities:

Definition (Linear equation)

A linear equation in one variable is an equation that can be put in the form $ax + b = 0$, where a and b are constants, and $a \neq 0$.

Example

$3x - 5 = 0$, $x = 5$, $\frac{1}{2}x + 8 = 0$ are linear equations.

- The solution of a linear equation $ax + b = 0$, where $a \neq 0$ is given by $x = -\frac{b}{a}$.
- If $a_1 \neq a_2$, the solution of an equation of the form $a_1x + b_1 = a_2x + b_2$ is given by $x = \frac{b_2 - b_1}{a_1 - a_2}$,
- If $a_1 = a_2$ the solution of $a_1x + b_1 = a_2x + b_2$ does not exist when $b_1 \neq b_2$, and has many solutions when $b_1 = b_2$.

To find the solution of a linear equations the following properties are important:

1. The addition property: If $a = b$ then $a + c = b + c$
2. The multiplication property: If $a = b$ then $ac = bc$

Example

Solve for x

(a) $820x = 10x + 30(50 - x)$

(b) $3(2x + 1) = 2(1 - 5x) + 6x + 11$

Example

Find the solution set of

(a) $\dfrac{8x + 3}{2} - 5(x + 2) = -3\left(x + \dfrac{5}{6}\right)$

(b) $5x - 2(x - 1) + 4 = 3(x + 2)$

(c) $6 + 3(1 - x) = 2(1 - 5x) + 7x$

Example

A computer discount store held an end of summer sale on two types of computers. They collected Birr 41,800 on the sale of 58 computers. If one type sold for Birr 600 and the other type sold for Birr 850, how many of each type were sold?

Definition (Linear inequalities)

A linear inequality is an inequality that can be put in the form $ax + b < 0$, where a, and b are constants with $a \neq 0$. ($<$ can be replaced with $>$, $\leq$, or $\geq$.)

Example

$2x - 7 > 9$, $5x \leq 4$, $1 - x \geq 10$ are examples of linear inequalities.

To solve the inequalities, we use properties of order relation.

Example

Find the solution set of

(a) $5x + 8(20 - x) \geq 2(x - 5)$

(b) $3x - 5(x + 2) \geq 0$

(c) $x - 4(x + 1) \geq -13 - (x - 2)$

Quadratic equations and inequalities

Definition (Quadratic equation)

A quadratic equation is an equation that can be put in the form $ax^2 + bx + c = 0$, where a, b and c are constants with $a \neq 0$.

Example

$3x^2 + x + 1 = 0$, $x^2 = 4$, $x^2 + 6x = 0$ are quadratic equations.

To solve a quadratic equation, the following properties can be used.

The zero-Product Rule If $a.b = 0$ then $a = 0$ or $b = 0$.

The square Root Theorem If $x^2 = d$, then $x = \pm\sqrt{d}$

Example

Solve the following equations

(a) $4x^2 + 10x = 6$ (b) $5x^2 - 6 = 8$ (c) $(x-2)^2 = 6$

The quadratic formula:

The solutions (roots) of a quadratic equation $ax^2 + bx + c = 0$, with $a \neq 0$ is given by

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Example

Solve the following equations using quadratic formula

(a) $x^2 - 8x = -6$ (b) $x^2 - 5x + 6 = 0$

(c) $x^2 + 4x + 4 = 0$ (d) $x^2 + x + 1 = 0$

Definition (Quadratic inequalities)

A quadratic inequality is an inequality that can be put in the form $ax^2 + bx + c < 0$, where a, b and c are constants with $a \neq 0$. ($<$ can be replaced with $>$, $\leq$, or $\geq$.)

Example

$2x^2 + 5x - 3 > 0$, $x^2 + 5x + 6 \leq 0$, $x - x^2 \geq 2$ are examples of linear inequalities.

Note that $a.b > 0$ if and only if both a and b are either positive or negative. We determine solution of the quadratic inequalities by examining the sign of the factors. After putting the inequality in the standard form, we will examine the sign of each factor of the expression for the various values of x. This process is called sign analysis.

Example

Solve the following quadratic inequalities

(a) $2x^2 + 5x - 3 > 0$ (b) $x^2 - 5x + 6 \leq 0$

(c) $x^2 - 2x - 2 < 0$

Cartesian Product

Definition

Suppose A and B are sets. The Cartesian product of A and B, denoted by $A \times B$, is the set which contains every ordered pair whose first coordinate is an element of A and second coordinate is an element of B. That is,

$$A \times B = \{(a, b) : a \in A and b \in B\}$$

Example

Let $A = \{1, 2\}$ and $B = \{x, y\}$. Find $A \times B$ and $B \times A$.

Definition (Equality of ordered pairs)

Two ordered pairs (a, b) and (c, d) are equal if and only if $a = c$ and $b = d$.

Definition (Relation)

Let A and B are sets, any subset of $A \times B$ is called a relation from A into B.

Example

1. Let $A = \{1, 3, 5, 7\}$ and $B = \{6, 8\}$. Let R be the relation "less than" from A into B. Then $R = \{(1, 6), (1, 8), (3, 6), (3, 8), (5, 6), (5, 8)\}$
2. Let $A = \{1, 2, 3, 4, 5\}$ and $B = \{a, b, c\}$. The following are relations from A into B.

   (i) $R_1 = \{(1, a)\}$

   (ii) $R_2 = \{(2, b), (3, b), (4, c), (5, a)\}$

Definition

Let R be a relation from A into B. Then

(a) the domain of R, denoted by $Dom(R)$, is the set of the first coordinates of the elements of R. That is

$$Dom(R) = \{a : (a, b) \in R\}$$

(ii) the range of R, denoted by $Range(R)$, is the set of the second coordinates of the elements of R. That is

$$Range(R) = \{b : (a, b) \in R\}$$

If R is the relation from A into B, then the set B is called the codomain of R.

Example

1. If $R = \{(1,6),(3,6),(3,8),(5,6),(5,8)\}$ is the relation from $A = \{1,3,5,7\}$ into $B = \{6,8,10\}$. Then find $Dom(R)$, $Range(R)$ and $Codomain(R)$.
2. Let R be a relation on $A = \{1,2,3,4,5\}$ defined by $R = \{(a,b) : a, b \in A, \text{a is a factor of b}\}$. Find the domain and range of R.

Remark

1. A relation R on a set A is called
   - **(a)** a universal relation if $R = A \times A$.
   - **(b)** identity relation if $R = \{(a,a) : a \in A\}$
   - **(c)** void or empty relation if $R = \emptyset$.
2. If R is a relation from A to B then the inverse of R, denoted by R^{-1}, is a relation from B to A and defined as:

$$R^{-1} = \{(b,a) : (a,b) \in R\}$$

Example

Let $R = \{(a, b) : a, b \in \mathbb{N}, a + 2b = 11\}$ be a relation on $\mathbb{N}$. Find
(a) $Dom(R)$ (b) $Range(R)$ (c) R^{-1}

Definition (Functions)

A function is a relation in which each element of the domain corresponds to exactly one element of the range.

Example

Determine whether the following relations are functions or not.

(a) $R = \{(5, -2), (3, 5), (3, 7), (5, 6), (5, 8)\}$

(b) $R = \{(2, 4), (3, 4), (6, 8)\}$

(c) $R = \{(-1, 1), (1, 1), (-2, 4), (2, 4), (-3, 9), (3, 9)\}$

Map or mapping, transformation and correspondence are synonyms for the word function. If f is a function and $(x, y) \in f$ then we say x is mapped to y.

Definition

A relation f from A into B is called a function from A into B, denoted by

$$f : A \to B \text{ or}$$

if and only if

(i) $Dom(f) = A$

(ii) No element of A is mapped by f to more than one element in B. That is, if $(x, y) \in f$ and $(x, z) \in f$, then $y = z$.

Remark

- If $(x, y) \in f$, then we write as $f(x) = y$. In this case, y is called the image of x, and x is called the pre-image of y under f.
- The symbol $f(x)$ is read as "f of x".

Example

1. Let $A = \{1, 2, 3, 4\}$ and $B = \{1, 6, 8, 11, 15\}$. which of the following are functions from A to B.
   - **(a)** f defined by $f(1) = 1, f(2) = 6, f(3) = 8, f(4) = 8$.
   - **(b)** f defined by $f(1) = 1, f(2) = 6, f(3) = 15$.
   - **(c)** f defined by $f(1) = 6, f(2) = 6, f(3) = 6, f(4) = 6$.
   - **(d)** f defined by $f(1) = 1, f(2) = 6, f(2) = 8, f(3) = 8, f(4) = 11$.
2. Is $f(x) = x^2$ a function from $\mathbb{R}$ to $\mathbb{R}$?

Domain, Codomain and range of a function

Definition

Let $f : A \to B$ be a function.

(a) The set A is called the domain of f.

(b) The set B is called the codomain of f.

(c) The set $\{f(x) : x \in A\}$ is called the range of f.

Example

Let $A = \{1, 2, 3\}$ and $B = \{1, 2, 3, \cdots 10\}$. Determine $Dom(f)$, $Range(f)$ and $Codomain(f)$, given that $f : A \to B$ is a function defined by $f(x) = x^2$.

Example

Determine the domain for the following functions.

(a) $f(x) = -3x + 5$ (b) $f(x) = \dfrac{2x}{3x - 5}$

(c) $f(x) = \sqrt{3x - x^2}$

Real Valued functions and their properties

- The function $f : A \to B$ is called a real valued function if $B \subseteq \mathbb{R}$, and in particular if A is also a subset of $\mathbb{R}$, then f is called real function.

Example

$f : \mathbb{R} \to \mathbb{R}$ defined by $f(x) = x^2$ is real function.

Operations of functions

Let $f(x)$ and $g(x)$ be functions.

1. $(f+g)(x) = f(x) + g(x)$ The sum of two functions.
2. $(f-g)(x) = f(x) - g(x)$ The difference of two functions.
3. $(f.g)(x) = f(x).g(x)$ The product of two functions.
4. $\left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)}$ The quotient of two functions(provided $g(x) \neq 0$).

The domain of $f+g$, $f-g$ and $f.g$ is $Dom(f) \cap Dom(g)$, whereas the domain of $\frac{f}{g}$ is $Dom(f) \cap Dom(g) - \{x : g(x) = 0\}$.

Example

1. Let $f(x) = 3x^2 + 2$ and $g(x) = 5x - 4$. Find each of the following, and its domain.
   (a) $(f+g)(x)$ (b) $(f-g)(x)$
   (c) $(f.g)(x)$ (d) $\left(\frac{f}{g}\right)(x)$
2. Let $f(x) = \sqrt[4]{x+1}$ and $g(x) = \sqrt{9-x^2}$. Find each of the following, and their domains.
   (a) $(f+g)(x)$ (b) $(f.g)(x)$
   (c) $\left(\frac{f}{g}\right)(x)$ (d) f^3

Definition (Composition of functions)

The composition of two functions $f(x)$ and $g(x)$, is denoted by fog and is defined by

$$(fog)(x) = f(g(x))$$

- The domain of fog consists of those $x's$ in the domain of g whose range values are in the domain of f.

Example

1. Let $f = \{(2, z), (3, q)\}$ and $g = \{(a, 2), (b, 3), (c, 5)\}$. Find fog and its domain.
2. Given $f(x) = 5x^2 - 3x + 2$ and $g(x) = 4x + 3$ find $fog(-2)$, $gof(-2)$, $fog(x)$ and $gof(x)$
3. If $f(x) = \dfrac{x}{x+1}$ and $g(x) = \dfrac{2}{x-1}$, find $fog(x)$, $gof(x)$ and their domains.

Definition (Equality of functions)

Two functions are said to be equal if and only if the following two conditions hold:

(i) The functions have the same domain;

(ii) Their functional values are equal at each element of the domain.

Example

The functions $f(x) = \dfrac{x^2 - 25}{x - 5}$ and $g(x) = x + 5$ are not equal. Why?

Types of functions

Definition

A function $f : A \to B$ is called one to one, often written $1-1$, if and only if for all $x_1, x_2 \in A$, $f(x_1) = f(x_2)$ implies $x_1 = x_2$. In other words, no two elements of A are mapped to one element of B.

Example

Let $A = \{1, 2, 3, 4\}$ and $B = \{1, 4, 7, 8\}$.

(i) $f : A \to B$ defined as
$f(1) = 1, f(2) = 4, f(3) = 4, f(4) = 8$ is not $1-1$.

(ii) $f : A \to B$ defined as
$f(1) = 4, f(2) = 7, f(3) = 1, f(4) = 8$ is $1-1$.

Definition

A function $f : A \to B$ is called an onto function(or f maps onto B) if every element of B is image of some element in A, i.e.

Definition

A function $f : A \to B$ is called an onto function(or f maps onto B) if every element of B is image of some element in A, i.e, $Range(f) = B$.

Example

$f : \mathbb{R} \to \mathbb{R}$ defined by $f(x) = 3x + 5$ is onto.
$f : \mathbb{R} \to \mathbb{R}$ defined by $f(x) = x^2$ is not onto.

Definition

A function $f : A \to B$ is called said to be a $1 - 1$ correspondence if f is both $1 - 1$ and on to.

Inverse of a function

Definition

Let f is a function whose domain is A, and whose codomain is B. Then f is invertible if there exists a function g with domain B and codomain A, with property:

$$f(x) = y \Leftrightarrow g(y) = x$$

The inverse of a function f is denoted by f^{-1} .

- Not all functions have inverse. A function f is invertible(has inverse) if it is one to one correspondence.

To find the inverse we will follow the steps below:

1. Interchange x and y in the equation $y = f(x)$
2. Solving the resulting equation for y , we will obtaining the inverse function. That is $y = f^{-1}(x)$.

Example

Find the inverse of the following functions, if exist.

(a) $f(x) = x^3$ $\qquad$ $f(x) = \dfrac{x}{x+2}$

Properties

If $f : A \to B$ and $g : B \to C$ are invertible,then

(a) $(f^{-1})^{-1} = f$

(b) $(f^{-1}of)(x) = x$

(c) $(fof^{-1})(y) = y$

(d) $(fog)^{-1} = g^{-1}of^{-1}$

Polynomial function

Definition

A polynomial function is a function of the form

$$p(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0,$$

where n is non negative integer, $a_n, a_{n-1}, \cdots, a_0$ are constants.

In a polynomial function $p(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0$, where $a_n \neq 0$,

- the constants $a_n, a_{n-1}, \cdots, a_0$ are called coefficients of the polynomial p,
- n is the degree of p,
- a_n is called the leading coefficient of p,
- $a_n x^n$ is the leading term, and a_0 is called the constant term of p.

Example

$f(x) = 2x^5 - 3x^4 - x^3 + 5x + 7$ is a polynomial function of degree 5. $g(x) = 2x^{-3} - 4x^2 + 10$ is not polynomial function (Why?)

Remark:

- A polynomial of degree 1 is called a linear function.
- A polynomial of degree 2 is called quadratic function.
- A polynomial of degree 3 is called a cubic function.

Properties

- The domain of the polynomial function is the set of real number.
- The graph of a polynomial is a smooth unbroken curve. The word smooth means that the graph does not have any sharp corners as turning points.

Definition

The zero (sometimes called roots) of a function f is (are) the value(s) of x such that $f(x) = 0$.

Example

$x = -2$ and $x = -3$ are the zeros of a quadratic function $f(x) = x^2 + 5x + 6$.

- If p is a polynomial of degree n , then it has at most n zeros.

Theorem (Division Algorithm)

Let $p(x)$ and $d(x)$ be polynomials with $d(x) \neq 0$, and with the degree of $d(x)$ less than or equal to the degree of $p(x)$. Then there are polynomials $q(x)$ and $r(x)$ such that
$p(x) = \underbrace{d(x)}_{\text{divisor}} \underbrace{q(x)}_{\text{quotient}} + \underbrace{r(x)}_{\text{remainder}}$ *, where either $r(x) = 0$ or the degree of $r(x)$ is less than degree $d(x)$.*

Example

Determine the quotient, and remainder obtained when $p(x) = x^4 - 1$ is divided by $d(x) = x^2 + 2x$

Theorem (The Remainder Theorem)

When a polynomial $p(x)$ of degree at least 1 is divided by $x - r$, then the remainder is $p(r)$.

Example

Find the remainder found in dividing $P(x) = x^3 - x^2 + 3x - 1$ by
(a) $x - 2$ (b) $x - 3$.

Theorem (The Factor Theorem)

$x - r$ is a factor of $p(x)$ if and only if $p(r) = 0$.

Example

Which of the following is a factor of $f(x) = x^3 + 3x^2 - 4x - 12$?
(a) $x - 2$ (b) $x + 3$ (b) $x + 2$

Theorem (Location theorem)

Let f be a polynomial function and a and b be real numbers such that $a < b$. If $f(a)f(b) < 0$, then there is at least one zero of f between a and b.

Example

Show that $2x^4 - 3x^2 + 5x - 7 = 0$ has atleast one real root on $[1, 2]$.

Theorem (Fundamental Theorem of Algebra)

If f be a polynomial function of degree $n > 0$ whose coefficients are complex numbers, then $p(x)$ has at least one zero in the complex number system.

Theorem (The linear Factorization Theorem)

If $p(x) = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0$ is a polynomial function of degree $n \geq 1$, then
$p(x) = a_n(x - r_1)(x - r_2)\cdots(x - r_n)$ where the r_i are complex numbers (possible real and not necessarily distinct).

Example

Factorize the following polynomial functions.

(a) $f(x) = x^3 - x6^2 - 16x$

(b) $f(x) = 3x^2 - 10x + 8$

(c) $f(x) = 2x^4 + 8x^3 + 10x^2$

Theorem (The Rational Root Theorem)

Suppose that $f(x) = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0$ *is a polynomial function of degree* $n \geq 1$ *with with integer coefficients. If* $\frac{p}{q}$ *is a rational root of* f *where* p *and* q *have no common factor other than* ± 1 *then* p *is a factor of* a_0 *and* q *is a factor of* a_n.

Example

Find all the zeros of the function $p(x) = 2x^3 + 3x^2 - 23x - 12$.

Rational Functions and their Graphs

Definition

A rational function is a function of the form $f(x) = \dfrac{n(x)}{d(x)}$, where $n(x)$ and $d(x)$ are polynomial functions and $d(x) \neq 0$.

Example

$f(x) = \dfrac{1}{x}$, $g(x) = \dfrac{2+x}{x^2+5x+6}$ and $h(x) = \dfrac{x^5+2x^3-x+1}{x^2+3x}$ are rational functions.

Note that the domain of the rational function $f(x) = \dfrac{n(x)}{d(x)}$ is $\{x : d(x) \neq 0\}$.

Example

Find the domain and the zero(s) of the following functions.

(a) $f(x) = \dfrac{3x-5}{x^2-x-12}$ (b) $f(x) = \dfrac{1}{x^3-4x}$

Informal definition of limits

x approaches a^+, x approaches of a^-, x approaches to $+\infty$, x approaches to $-\infty$.

Graph of rational functions

To sketch the graph of a function, the following information are important.

Domain, intercepts, asymptotes, Maximum and minimum values, etc. Define the following!

- Domain?
- Intercept: the x intercept, y intercept.
- Asymptotes: Vertical asymptote(s), Horizontal asymptote(s), Oblique asymptote(s).

Example

Sketch the graphs of

(a) $f(x) = \dfrac{1}{x}$ (b) $g(x) = \dfrac{1}{(x-a)^n}$, where n is positive integer.

Example

Find the Horizontal asymptote(s) of

(a) $f(x) = \dfrac{3x^3 - x^2 + 5x - 9}{2x^4 - 5x^3 + x^2 + x - 6}$

(b) $g(x) = \dfrac{3x^3 - x^2 + 5x - 9}{2x^3 + 4x^2 + 5x + 2}$

(c) $h(x) = \dfrac{-x^2 + 4x + 3}{5x^3 - 3x^2 + 7x + 8}$

Example (Exercise)

Sketch the graphs of

(a) $f(x) = \dfrac{x+2}{x-1}$ (b) $g(x) = \dfrac{x^2+3x+2}{x^2-1}$

Exponential function

Definition

For a natural number n and a real number x, the nth power of x is defined as

$$x^n = \underbrace{x.x.\cdots.x}_{\text{n times}}$$

In the expression x^n, x is called the base and n is called the exponent.

Example

$2^5 = 2 \times 2 \times 2 \times 2 \times 2 = 32$

$$\left(-\frac{1}{3}\right)^3 = -\frac{1}{3} \times -\frac{1}{3} \times -\frac{1}{3} = \frac{-1 \times -1 \times -1}{3 \times 3 \times 3} = -\frac{1}{27}$$

Definition (Zero and Negative exponent)

For $x \neq 0$, $x^0 = 0$ and $x^{-n} = \dfrac{1}{x^n}$.

Note that 0^0 is undefined.

Definition (Exponential function)

A function of the form $f(x) = b^x$, where $b > 0$ and $b \neq 1$, is called an exponential function with base b.

Example

$f(x) = 2^x$, $g(x) = \left(\frac{1}{3}\right)^x$ and $h(x) = e^x$ are exponential functions.

Properties of exponential function

- the domain of the exponential function is the set of real number, and the range is the set of positive real numbers.
- The y -intercept is 1.
- $y = 0$ is a horizontal asymptote.
- The exponential function is 1 -1.

Example

Sketch the graphs of (a) $f(x) = 3^x$ (b) $g(x) = \left(\frac{1}{3}\right)^x$

Example (Exercise)

Sketch the graphs of (a) $f(x) = 3^x + 1$ (b) $g(x) = 3^{x+1}$
(c) $g(x) = -9^{-x} + 3$

Logarithm function

Because the exponential function is 1-1, it has an inverse function.
Question: What is the inverse of $f(x) = b^x$, where $b > 0$ and $b \neq 1$?

Example

Determine the inverse of $f(x) = 3^x$.

Note: For $b > 0$ and $b \neq 1$, $x = b^y \Leftrightarrow y = \log_b x$.

Definition (Logarithm function)

A function of the form $f(x) = \log_b x$, where $b > 0$ and $b \neq 1$, is called a logarithm function with base b.

Example

$f(x) = \log_2 x$, $g(x) = \log_{\frac{1}{2}} x$ are logarithm functions.

Properties of logarithm function

- the domain of the logarithm function is the set of positive real number, and the range is the set of real numbers.
- The x -intercept is 1.
- $x = 0$ is a vertical asymptote.
- The logarithm function is 1 -1.

Example

Sketch the graphs of (a) $f(x) = \log_2 x$ (b) $g(x) = \log_{\frac{1}{2}} x$

Definition

Common Logarithm: $f(x) = \log_{10} x$ is called common logarithm function. We simply write as $f(x) = \log x$.

Natural Logarithm: $f(x) = \log_e x$ is called natural logarithm function. We simply write as $f(x) = \ln x$.

Mathematics for Social Sciences

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Department of Mathematics
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2021/22

Out lines

1. **Matrix, Determinant and Systems of linear equations**
   - Definitions and Examples
   - Matrix algebra(operations)
   - Determinant and it's Properties
   - Systems of Linear Equations

Matrix

Definition

A matrix with m rows and n columns is a rectangular array of numbers of the form

$$\begin{pmatrix} a_{11} & a_{12} & . & . & . & a_{1n} \\ a_{21} & a_{22} & . & . & . & a_{2n} \\ . & . & & & & . \\ . & . & & & & . \\ . & . & & & & . \\ a_{m1} & a_{m2} & . & . & . & a_{mn} \end{pmatrix},$$

- The size or order of a matrix with m rows and n columns is $m \times n$ (and read it as m by n).

- We often use capital letters to denote matrices, and lower case letters to denote its elements. In short, we write a matrix as $A = (a_{ij})_{m\times n}$ or $A = (a_{ij})_{m,n}$ or simply $A = (a_{ij})$ if the order is not important to emphasis.
- The element a_{ij} of a matrix $A = (a_{ij})_{m\times n}$ is located in the ith row and the jth column.

Example

$A = \begin{pmatrix} -4 & 3 & \frac{1}{3} \\ 1 & -1 & 6 \end{pmatrix}$ *and* $B = \begin{pmatrix} 1 & -2 \\ 2 & 4 \\ -4 & 16 \end{pmatrix}$ *are a* 2×3 *and a* 3×2 *matrices respectively*

Definition

A matrix with a single column (or an $n \times 1$ matrix) is called a column vector(or column matrix). A matrix with single row(or $1 \times n$ matrix) is called a row vector(or row matrix).

Example

$A = \begin{pmatrix} 5 \\ 6 \\ 11 \end{pmatrix}$ *is a column vector(matrix) and* $B = \begin{pmatrix} 1 & 2 & 3 \end{pmatrix}$ *is a row vector(matrix).*

Definition

An $m \times n$ matrix $A = (a_{ij})$ is said to be the zero (null) matrix if $a_{ij} = 0$ for all i, j.

Example

$\mathbf{O} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$ *is a* 2×2 *zero matrix,* $\mathbf{O} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{pmatrix}$ *is a* 3×2 *zero matrix.*

Definition

A matrix $A = (a_{ij})_{m \times n}$ is called a square matrix of order n(or an n square matrix) if m=n.

Example

$$A = \begin{pmatrix} 5 & 4 & 1 \\ -1 & 0 & 2 \\ 0 & 3 & -6 \end{pmatrix} \textit{ is a square matrix.}$$

Definition

The entries $a_{11}, a_{22}, a_{33}, ...a_{nn}$ are called the main (principal) diagonal elements of a square matix $(a_{ij})_n$.

Types of square matrices

(a) Triangular matrix:
A square matrix $A = (a_{ij})_n$ is said to be a

- **(i)** lower triangular if all the entries above the main diagonal are 0, or $a_{ij} = 0$ for all $i < j$
- **(ii)** upper triangular if all the entries below the main diagonal are all 0, or $a_{ij} = 0$ for all $i > j$.
- **(iii)** triangular if it is either upper or lower triangular matrix.

(b) Diagonal matrix: A square matrix D is called a diagonal matrix if all the elements except the principal diagonal elements are zero.

Example

content...

(c) Scalar matrix:
A diagonal matrix D is called a scalar matrix if the main diagonal elements are equal.

(d) Identity matrix:
A scalar matrix whose all of its diagonal elements are 1, is an identity matrix.

Notation: An identity matrix of order n is denoted by I_n

Equality of Matrices

Definition

Two matrices A and B are said to be equal(written as $A = B$) if they have the same size and their corresponding elements are equal.

Example

$A = \begin{pmatrix} 1 & 2 & -1 \\ 2 & -3 & 4 \\ 0 & -4 & 5 \end{pmatrix}$ *and* $B = \begin{pmatrix} 1 & 2 & w \\ 2 & x & 4 \\ y & -4 & z \end{pmatrix}$ *are equal if and only if* $w = -1, x = -3, y = 0, z = 5$.

Matrix algebra

Definition (Scalar Multiplication)

Let $A = (a_{ij})_{m\times n}$ be a matrix, and α is a scalar (a number). The product of the scalar α with matrix A, denoted by αA, is defined by

$$\alpha A = (\alpha a_{ij})_{m\times n}$$

Example

Let $A = \begin{pmatrix} 3 & -2 & 1 \\ 0 & 4 & 2 \end{pmatrix}$. Find $3A$.

Addition of Matrices

Definition

Let $A = (a_{ij})_{m\times n}$ and $B = (b_{ij})_{m\times n}$ be two matrices. The sum of A and B, denoted by $A + B$ is defined by

$$A + B = (a_{ij} + b_{ij})_{m\times n}$$

Example

Let $A = \begin{pmatrix} -3 & 5 & -1 \\ 1 & -1 & 7 \end{pmatrix}$ and $B = \begin{pmatrix} 4 & -4 & -6 \\ 5 & 11 & 4 \end{pmatrix}$. Find $A + B$ and $B + A$

Properties

Theorem

Let A, B, C be $m \times n$ matrices and α, β be constants. Then

(a) $+$ *is commutative*

(b) $+$ *is associative*

(c) *Existence of additive identity*

(d) *Existence of an additive inverse matrix*

(e) *scalar multiplication is distributive over addition.*

(f) $(\alpha + \beta)A = \alpha A + \beta A$

(g) $(\alpha\beta)A = \alpha(\beta A) = \beta(\alpha A)$

(h) $0A = \mathbf{O}$, $\alpha\mathbf{O} = \mathbf{O}$

Multiplication of Matrices

Definition

Let $A = (a_{ij})_{m\times n}$ and $B = (b_{ij})_{n\times p}$ matrices. Then the matrix product AB is defined as $C = (c_{ij})_{m\times p}$ with entries

$$\begin{aligned} c_{ij} &= a_{i1}b_{1j} + a_{i2}b_{2j} + a_{i3}b_{3j} + \ldots + a_{in}b_{nj} \\ &= \sum_{k=1}^{n} a_{ik}b_{kj}, \text{ for } 1 \le i \le m, 1 \le j \le p \end{aligned}$$

Example

Let $A = \begin{pmatrix} -1 & 3 \\ 1 & -2 \\ 0 & 4 \end{pmatrix}$, $B = \begin{pmatrix} 0 & -5 \\ -3 & 2 \end{pmatrix}$,

$C = \begin{pmatrix} 1 & -3 & 4 \\ -2 & 0 & 1 \end{pmatrix}$. *Then find AB, BA, AC and CA (if exist).*

Matrix multiplication is not commutative.

Theorem

Let A,B and C be matrices for which all products below make sense. Then

(a) *Matrix multiplication is associative.*
That is, $A(BC) = (AB)C$

(b) $A(B + C) = AB + AC$, $(A + B)C) = AC + BC$

(c) $AI = A$ *and* $IA = A$

(d) $\alpha(AB) = (\alpha A)B = A(\alpha B)$

(e) $A\mathbf{O} = \mathbf{O}$

Transpose of matrix

Definition

The transpose of a matrix $A = (a_{ij})_{m\times n}$, is the matrix $A^t = (a_{ji})_{n\times m}$.

Example

Find A^t for $A = \begin{pmatrix} 5 & -4 & 7 \\ 0 & 6 & -10 \\ -4 & 3 & 1 \end{pmatrix}$.

Theorem (Laws of transpose)

If A and B are matrices of the appropriate sizes, α, β are constants, then

(a) $(A^t)^t = A$

(b) $(\alpha A + \beta B)^t = \alpha A^t + \beta B^t$

(c) $(AB)^t = B^t A^t$

Symmetric and Skew Symmetric Matrices

Definition

A square matrix A is called

(a) symmetric if $A^t = A$.

(b) skew symmetric if $A^t = -A$.

Example

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Elementary row operations

The following operations performed on a matrix A are called elementray operations.

(i) Interchanging two rows ($R_i \leftrightarrow R_j$)

(ii) Multiplying a row by a non zero constant ($kR_i \rightarrow R_i$)

(iii) Adding a multiple of entries of a row to the corresponding entries of one another row ($R_j + kR_i \rightarrow R_j$).

Example

Using elementary row operations, transform the matrix $A = \begin{pmatrix} 3 & 12 & 6 \\ 1 & 1 & -1 \\ 1 & 2 & 3 \end{pmatrix}$ to an upper triangular matrix.

Remark: Two matrices are said to be row equivalent if one can be obtained from the other by a sequence of elementary row operations.

Definition (Row Echelon and Row Reduced Echelon Forms)

A matrix $A = (a_{ij})_{m \times n}$ is a row echelon form of a matrix if

(a) all the zero rows (if any) are at the bottom of rows of A

(b) the first non zero entry in each non zero row is 1. We call this leading 1.

(c) each leading 1 is to the right of the leading 1 element in the proceeding rows.

Example

content...

Definition

A matrix $A = (a_{ij})_{m\times n}$ is a row reduced echelon form of a matrix if

(a) A is a row echelon form

(b) If a column contains a leading 1, then all other entries in that column are zero

Example

content...

Remark: We apply a sequence of elementary operations on a matrix A to obtain its row echelon or row reduced echelon form.

Example

Find a row echelon and row reduced forms of the matrix

$$A = \begin{pmatrix} -1 & 2 & -5 \\ 2 & -1 & 6 \\ 1 & 1 & 3 \end{pmatrix}$$

Rank of a matrix

Definition

The rank of a matrix A, denoted by $rank(A)$, is the number of non zero rows(columns) of its equivalent row echelon form of a matrix.

Example

Find the rank of the following matrices.

(a) $A = \begin{pmatrix} 1 & 1 & 2 \\ 2 & 2 & 5 \\ 3 & 3 & 2 \end{pmatrix}$ **(b)** $B = \begin{pmatrix} 1 & 2 \\ -2 & 4 \\ 0 & -3 \end{pmatrix}$

Inverse of a Matrix

Definition

Let A be a square matrix of size n. The matrix A is said to be **invertible** (non-singular) if there exist a square matrix B of size n such that $AB = BA = I_n$.

Example

Verify that $B = \begin{pmatrix} \frac{2}{5} & -\frac{3}{5} \\ \frac{1}{5} & \frac{1}{5} \end{pmatrix}$ is an inverse of $A = \begin{pmatrix} 1 & 3 \\ -1 & 2 \end{pmatrix}$.

Notation: The inverse of an invertible matrix A is denoted as A^{-1}.

Determinant and it's Properties

The determinant is a function which assigns to each square matrix $A = (a_{ij})_n$ a number, and it is denoted by det(A) or $|A|$ or Δ.

A. Determinant of square matrix of order 1 and order 2

(i) For a 1×1 matrix $A = (a_{11})$, $det(A) = a_{11}$.

(ii) For a 2×2 matrix $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$,

$$det(A) = \begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc.$$

Example

Find the determinants of the following matrices.

(a) $A = \begin{pmatrix} -2 \end{pmatrix}$ **(b)** $A = \begin{pmatrix} -3 & 5 \\ 1 & 4 \end{pmatrix}$ **(c)** $A = \begin{pmatrix} 1 & 2 \\ 3 & 6 \end{pmatrix}$

Minors and Cofactors

Definition

Let $A = (a_{ij})_n$ be a matrix and A_{ij} be an $n-1 \times n-1$ sub matrix of A obtained from A by deleting the ith row and the jth column.

(a) The minor of a_{ij},denoted by M_{ij} is the determinant of A_{ij}.

(b) the cofactor of a_{ij}, denoted by C_{ij} is defined as $C_{ij} = (-1)^{i+j} M_{ij}$

(c) Matrix of Cofactors of a matrix, denoted by C is a matrix $C = (C_{ij})_n$.

Example

Let $A = \begin{pmatrix} 1 & -1 & 2 \\ -2 & 0 & 5 \\ -3 & 4 & 3 \end{pmatrix}$. Find $M_{11}, M_{12}, M_{22}, M_{32}$, $C_{11}, C_{12}, C_{22}, C_{32}$

B. Determinant of square matrix of order n, $n \geq 3$.

Definition

For a square matrix $A = (a_{ij})_n$,

(a) $det(A) = a_{i1}C_{i1} + a_{i2}C_{i2} + ... + a_{in}C_{in} = \sum_{k=1}^{n} a_{ik}C_{ik}$, where, $1 \leq i \leq n$.
(Cofactor expansion along the ith row)

(b) $det(A) = a_{1j}C_{1j} + a_{2j}C_{2j} + ... + a_{nj}C_{nj} = \sum_{k=1}^{n} a_{kj}C_{kj}$, where, $1 \leq j \leq n$.
(Cofactor expansion along the jth column).

Example

content...

Properties of determinant

(a) If a matrix B results from a square matrix A by interchanging two different rows(columns) of A, then

$$det(B) = -det(A)$$

(b) If B is obtained from A by multiplying a row(column) of A by a real number k, then

$$det(B) = kdet(A)$$

.

(c) If A is a square matrix of order n, $det(kA) = k^n det(A)$ for some constant k.

(d) If B is obtained from A by adding to each element of the rth row(column) of A, k times the corresponding element of the sth row(column), $r \neq s$ of A, then

$$det(A) = det(B)$$

(e) If A is a square matrix, $det(A) = det(A^t)$

(f) If two rows(columns) of a square matrix A are proportional then

$$det(A) = 0$$

(g) If a row(column) of A consists entirely of zeros, then $det(A) = 0$.

(h) The determinant of a triangular matrix is the product of the elements on the main diagonal.

(i) If A and B are square matrices of the same order,

$$det(AB) = det(A)det(B)$$

(j) If A is invertible, then $det(A^{-1}) = \frac{1}{det(A)}$.

Adjoint of a matrix

Definition

Let A be $n \times n$ matrix and C_{ij} denote the cofactor of a_{ij}. The adjoint of A, denoted by $adj(A)$, is the transpose of the matrix of cofactors, $C = (C_{ij})_n$.

Theorem (Determinant test for invertiblity)

A square matrix A is invertible if and only if $detA \neq 0$.

Corollary

If A is an invertible matrix, then $A^{-1} = \frac{1}{det(A)} adj(A)$

Example

Find the inverse of each of the following matrices(if exist).

(a) $A = \begin{pmatrix} 2 & 3 & -4 \\ 0 & -4 & 2 \\ 1 & -1 & 5 \end{pmatrix}$ **(b)** $B = \begin{pmatrix} 1 & 2 & 2 \\ 1 & 3 & 1 \\ 1 & 1 & 3 \end{pmatrix}$

Systems of Linear Equations

Definition

A system of m linear equations in n unknowns $x_1, x_2, ...x_n$ is a set of linear equations given by

$$\begin{aligned}
a_{11}x_1 + a_{12}x_2 + ... + a_{1n}x_n &= b_1 \\
a_{21}x_1 + a_{22}x_2 + ... + a_{2n}x_n &= b_2 \\
. \qquad\qquad\qquad\qquad\qquad\qquad &\quad . \\
. \qquad\qquad\qquad\qquad\qquad\qquad &\quad . \\
. \qquad\qquad\qquad\qquad\qquad\qquad &\quad . \\
a_{m1}x_1 + a_{m2}x_2 + ... + a_{mn}x_n &= b_m,
\end{aligned}$$

Using a matrix relation we write as

$$AX = B \tag{1}$$

where $A = (a_{ij})_{m\times n}$, $X = \begin{pmatrix} x_1 & x_2 & . & . & . & x_n \end{pmatrix}^t$, and $B = \begin{pmatrix} b_1 & b_2 & . & . & . & b_m \end{pmatrix}^t$

- A matrix A of the system (1) is called **the coefficient matrix**, and the matrix obtained by adjoining the column vector B at the end to the matrix A is called **the augmented matrix** of the system, denoted by $M = (A|B)$ or $M = (A : B)$.
- If $B = \mathbf{O}$, the system (1) is called **homogenous**; otherwise, it is called **non homogenous**.

Example

Given a system

$$\begin{aligned} 2x_1 + x_2 - 5x_3 &= 5 \\ x_2 + x_2 - 4x_3 &= 0 \\ -5x_1 - 6x_3 &= 12 \end{aligned}$$

Determine

(a) *The coefficient and the augmented matrices.*

(b) *The matrix form of the system.*

For the system of linear equations in (1), precisely one of the statements below is true:

1. It admits a unique Solution: There is one and only one vector $x = (x_1, x_2, ...x_n)$ that satisfies all the m-equations simultaneously (the system is consistent).
2. It has infinitely Many Solutions: There are infinitely many different values of x that satisfy all the m-equations simultaneously (the system is said to be consistent).
3. Has no Solution: There is no vector x that satisfies all equations simultaneously, or the solution set is empty (the system is said to be inconsistent).

Solving systems of systems of linear equations

Matrix Methods: Gaussian Elimination, Gauss-Jordan method, Cramer's rule, Inverse Method.

1. Gaussian Elimination:

The method of solving a system $AX = B$ by transforming the augmented matrix $M = (A|B)$ to its row echelon form using elementary row operations is called Gaussian Elimination.

2. Gauss-Jordan Method:

The method of solving a system $AX = B$ by transforming the augmented matrix M to row reduced echelon form of a matrix of the system using elementary row operations is called Gauss-Jordan reduction.

Example

Using Gaussian elimination method, solve the system of linear equations.

$$x + 2y + z = 3$$
$$2x + 3y - z = -6$$
$$3x - 2y - 4z = -2$$

Theorem

Let A be an invertible $n \times n$ matrix, A_{x_i} denote the matrix obtained from A by replacing the ith column of A with the n-vector B. If $|A| \neq 0$, the linear system $AX = B$, where $X = \begin{pmatrix} x_1 & x_2 & . & . & . & x_n \end{pmatrix}^t$ has a unique solution given by $x_i = \frac{det(A_{x_i})}{det(A)}$ where $i = 1, 2, 3, ..., n$

3.Cramer's rule:

Example

Using Cramer's rule, solve the system of linear equations.

$$2x_1 - x_2 = 0$$
$$-x_1 + 2x_2 - x_3 = 0$$
$$-x_2 + x_3 = 1$$

4. Inverse method:

Theorem

If A is an invertible matrix, the linear system $AX = B$ has a unique solution, which is given by

$$X = A^{-1}B$$